

SYSTEM MODULES

1. Front-End Module

- Provides two separate login interfaces:
 - Toll Operator login
 - Higher Authority (NHAI) login
- Toll Operator can:
 - Upload recorded video or use live camera feed
 - View detected vehicle details
 - View real-time alerts
- NHAI can:
 - View alerts from all toll plazas
 - Add vehicles to the database
 - Edit vehicle details
 - Remove vehicles from the database
- Implements role-based access control.

2. Back-End Module

- Handles all requests from the front end.
- Authenticates users and manages access roles.
- Controls data flow between AI module, database, and alert system.
- Performs vehicle verification logic.
- Sends real-time alert data to the traffic control room.

3. AI / Vehicle Detection Module

- Processes live and recorded video input.
- Detects vehicles from video frames.
- Extracts vehicle number plate details

- Sends detected vehicle data to the back end in real time.

4. Database Module

- Stores authorized / reference vehicle data (managed by NHAI).
- Stores recently detected vehicle data for verification.
- Supports fast data comparison.
- Used for real-time processing, not long-term alert history.

5. Alert System Module

- Generates alerts when suspicious or matched vehicles are detected.
- Uses the latest verified vehicle data.
- Sends alerts to:
 - Toll operator interface
 - Nearby traffic control room
- Operates in real time.